

DSA Daily MCQ 1

Total points 15/15 ?

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✓ Which of the following case does not exist in complexity theory? *

1/1

- ☐ Best case
- ☐ Worst case
- ☐ Average case
- ☒ Null case



✓ The complexity of linear search algorithm is _____ * 1/1

- ☒ $O(n)$
- ☐ $O(\log n)$
- ☐ $O(n^2)$
- ☐ $O(n \log n)$



✓ The complexity of Binary search algorithm is _____ * 1/1

- ☐ $O(n)$
- ☒ $O(\log n)$
- ☐ $O(n^2)$
- ☐ $O(n \log n)$



✓ The Worst case occur in linear search algorithm when _____ * 1/1

- ☐ Item is somewhere in the middle of the array
- ☐ Item is not in the array at all
- ☐ Item is the last element in the array
- ☒ Item is the last element in the array or is not there at all



✓ The complexity of Fibonacci series is _____ *

1/1

- ☒ $O(2^n)$
- ☐ $O(\log n)$
- ☐ $O(n^2)$
- ☐ $O(n \log n)$



✓ What is the time complexity of following code: *

1/1

```
int a = 0;
for (i = 0; i < N; i++)
{
    for (j = N; j > i; j--)
    {
        a = a + i + j;
    }
}
```

- ☐ $O(N)$
- ☐ $O(N \cdot \log(N))$
- ☐ $O(N * \text{Sqrt}(N))$
- ☒ $O(N \cdot N)$



✓ What is the time complexity of following code: *

1/1

```
int i, j, k = 0;
for (i = n / 2; i <= n; i++)
{
    for (j = 2; j <= n; j = j * 2)
    {
        k = k + n / 2;
    }
}
```

- ☐ $O(n)$
- ☒ $O(n \log n)$
- ☐ $O(n^2)$
- ☐ $O(n^2 \log n)$



✓ What does it mean when we say that an algorithm X is asymptotically more efficient than Y?

*1/1

- ☐ X will always be a better choice for small inputs
- ☒ X will always be a better choice for large inputs
- ☐ Y will always be a better choice for small inputs
- ☐ X will always be a better choice for all inputs



✓ To measure Time complexity of an algorithm Big O notation is used which: *1/1

- ☐ describes limiting behaviour of the function
- ☐ characterises a function based on growth of function
- ☐ upper bound on growth rate of the function
- ☒ all of the mentioned



✓ If for an algorithm time complexity is given by $O(1)$ then complexity of it is: *1/1

- ☒ constant
- ☐ polynomial
- ☐ exponential
- ☐ none of the mentioned



✓ If for an algorithm time complexity is given by $O(\log_2 n)$ then complexity will be : *1/1

- ☐ constant
- ☐ polynomial
- ☐ exponential
- ☒ none of the mentioned



✓ If for an algorithm time complexity is given by $O(n)$ then complexity of it is: *1/1

- ☐ constant
- ☒ linear
- ☐ exponential
- ☐ none of the mentioned



✓ if for an algorithm time complexity is given by $O(n^2)$ then complexity will be: *1/1

- ☐ constant
- ☒ quadratic
- ☐ exponential
- ☐ none of the mentioned



✓ If for an algorithm time complexity is given by $O((3/2)^n)$ then complexity will be: *1/1

- ☐ constant
- ☐ quadratic
- ☒ exponential
- ☐ none of the mentioned



✓ Two main measures for the efficiency of an algorithm are *

1/1

- ☒ Time and space
- ☐ Processor and memory
- ☐ Complexity and capacity
- ☐ Data and space



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